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#include <Adafruit_Fingerprint.h>
#include <SoftwareSerial.h>

// Define the pins connected to the fingerprint sensor
SoftwareSerial mySerial(2, 3); // RX, TX

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

void setup() {
  Serial.begin(9600);
  Serial.println("Fingerprint sensor initializing...");

  // Initialize fingerprint sensor
  finger.begin(57600);
  if (finger.verifyPassword()) {
    Serial.println("Fingerprint sensor found!");
  } else {
    Serial.println("Fingerprint sensor not detected!");
    while (1); // Stop if sensor is not detected
  }

  Serial.println("Ready to enroll a fingerprint or verify!");
}

void loop() {
  Serial.println("\nChoose an option:");
  Serial.println("1. Enroll a new fingerprint");
  Serial.println("2. Verify a fingerprint");
  Serial.println("Enter the option number:");

  while (Serial.available() == 0) {
    // Wait for input
  }

  int option = Serial.parseInt();
  if (option == 1) {
    enrollFingerprint();
  } else if (option == 2) {
    verifyFingerprint();
  } else {
    Serial.println("Invalid option. Try again.");
  }
}

void enrollFingerprint() {
  int id = 1; // Set an ID for this fingerprint. Use a unique ID for each
  fingerprint.

  Serial.print("Enrolling fingerprint with ID #");
  Serial.println(id);
}

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int p = -1;
Serial.println("Place your finger on the sensor.");
while (p != FINGERPRINT_OK) {
    p = finger.getImage();
    if (p == FINGERPRINT_OK) {
        Serial.println("Image taken");
    } else if (p == FINGERPRINT_NOFINGER) {
        Serial.print(".");
    } else {
        Serial.println("Error taking image");
        return;
    }
    delay(100);
}

p = finger.image2Tz(1);
if (p != FINGERPRINT_OK) {
    Serial.println("Error converting image.");
    return;
}

Serial.println("Remove your finger and place it again.");
delay(2000);
while (p != FINGERPRINT_OK) {
    p = finger.getImage();
    if (p == FINGERPRINT_OK) {
        Serial.println("Image taken");
    } else if (p == FINGERPRINT_NOFINGER) {
        Serial.print(".");
    } else {
        Serial.println("Error taking image");
        return;
    }
    delay(100);
}

p = finger.image2Tz(2);
if (p != FINGERPRINT_OK) {
    Serial.println("Error converting image.");
    return;
}

// Create a model for the fingerprint
p = finger.createModel();
if (p == FINGERPRINT_OK) {
    Serial.println("Fingerprint matched!");
} else {
    Serial.println("Error creating model.");
    return;
}

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    }

    // Store the model in flash memory
    p = finger.storeModel(id);
    if (p == FINGERPRINT_OK) {
        Serial.println("Fingerprint enrolled successfully!");
    } else {
        Serial.println("Error storing fingerprint.");
    }
}

void verifyFingerprint() {
    Serial.println("Place your finger on the sensor to verify.");
    int p = finger.getImage();
    if (p != FINGERPRINT_OK) {
        Serial.println("No finger detected.");
        return;
    }

    p = finger.image2Tz();
    if (p != FINGERPRINT_OK) {
        Serial.println("Error converting image.");
        return;
    }

    p = finger.fingerFastSearch();
    if (p == FINGERPRINT_OK) {
        Serial.print("Fingerprint verified! ID #");
        Serial.println(finger.fingerID);
    } else {
        Serial.println("Fingerprint not recognized.");
    }
}

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